

No. of Functional Features Included in the Solution

Date	13 July 2026
Team ID	XXXXXX
Project Name	Comprehensive Measure of Well-being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Missing Value Imputation	Cleans raw dataset variables by resolving trailing spaces and empty cells automatically.	Data Preprocessing Pipeline	Done	Core
2	Feature Scaling	Standardizes various metrics to prepare uniform arrays for regression logic computation.	Data Preprocessing Pipeline	Done	Core
3	Model Weight Serialization	Saves the fully trained machine learning parameter states securely into a file-based binary (HDI.pkl).	Machine Learning Core	Done	Core
4	Parameter Form Fields	Provides interactive form fields to safely capture socio-economic user stats like schooling and life expectancy.	User Interface View	Done	Core
5	Live Routing Network	Processes HTTP POST form requests, parsing inputs directly into model pipelines instantly.	Flask Backend Gateway	Done	Core
6	Score Boundary Clamp	Mathematically ensures all generated calculation metrics map strictly within a logical range of 0.0 to 1.0.	Machine Learning Core	Done	Core

7	Automated Report Export	Generates a clean, downloadable summary report outlining calculated regional well-being scores.	Reporting Module	Pending	Nice-to-Have
S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
8	Multi-Model Dynamic Toggle	Allows frontend users to seamlessly switch prediction calculations between alternative ML models.	Core Integration	Pending	Nice-to-Have

Feature Summary:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Parameter Form Fields
2	Backend / Logic	4	Missing Value Imputation, Feature Scaling, Score Boundary Clamp, Multi-Model Dynamic Toggle
3	Database / Storage	1	Model Weight Serialization
4	API / Integration	2	Live Routing Network, Automated Report Export
5	Security / Authentication	0	-
Metric			Count / Value
Total Features Planned			8
Total Features Implemented			6
Core / Must-Have Features			6

Additional / Nice-to-Have Features	2
Features Tested & Verified	6

Feature Category Breakdown: